spinmate

User Guide

Version 1.0 · free and open source (MIT) · github.com/Michaellianeris/sublattice-agent

An AI agent for spintronics simulation. Describe the setup in plain language or provide a parameter file. SpinMate validates the input with the solver's own parser, runs two-sublattice macrospin simulations, and explains the results from the actual magnetization dynamics, with figures and reports ready to share.

At a glance

- Chat-driven runs, sweeps and threshold searches on the original macrospin (LLG) code, unchanged
- Answers grounded in computed metrics: switches, settling, frequencies
- Claude and OpenAI behind four personas, chosen per request by capability and price
- Figures (IEEE, Nature), reports with LaTeX math, a searchable run catalog
- Runs locally: Docker or a Windows app; your keys, your data

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1 Getting started

Docker (Linux, macOS, Windows/WSL)

```
cp .env.example .env      # add
ANTHROPIC_API_KEY and/or OPENAI_API_KEY
docker compose up -d      # then open
http://localhost:8421
```

Windows desktop app

Run `SpinMate.exe`; it starts the server and opens the browser. Runs are kept in `%LOCALAPPDATA%\SpinMate\data` and the catalog is SQLite.

First start

1. **Athena**, the owl, asks your name and, optionally, your email. The email creates an account and, if a mail server is set up, brings this guide as a welcome email; otherwise she gives you the download link.
2. **API key:** Settings (gear) → paste an Anthropic and/or OpenAI key. A pasted key stays in that browser tab; a key in `.env` serves everyone on the server.
3. **Ask** in the message box, or pick a card under *Find a skill*.

No key? *Run setup* (play icon) runs a preset or your parameter file directly; only the chat needs a model.

2 Your first simulation

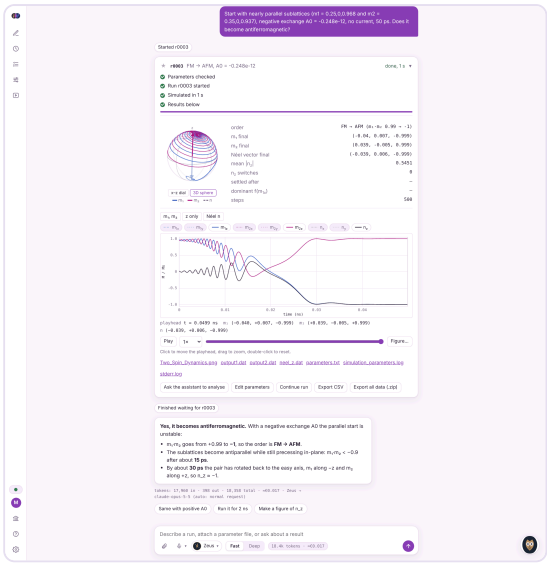
Describe the starting state, exchange, fields, current or temperature, and the duration (SI units, currents in A/cm²).

> *Start with nearly parallel sublattices (m1 = 0.25,0,0.968 and m2 = 0.35,0,0.937), A0 = −0.248e-12, 50 ps. Does it become antiferromagnetic?*

> *Start antiferromagnetic with a positive exchange A0 = 0.248e-12, 120 ps. Does it end ferromagnetic?*

> *Apply a DC spin-orbit current of 100e6 A/cm² for 2 ns to the default AFM state. What happens?*

> *Heat a FeRh layer through its transition: positive A0, 370 K ± 60 K at 0.25 GHz, 2 ns.*



Question, run card and answer with next steps.

The parameters are checked with the solver's own parser before anything runs; a suspicious value (for example `A0 = 0.248` without `e-12`) is pointed out instead of run. Large requests (many runs or long runtimes) wait for your **Run it** button. The answer ends with suggested next steps as buttons.

What happens behind one question

1. The router picks a model for the persona and the request.
2. The model calls `validate_parameters`, then `start_simulation`.
3. The run queue starts the solver in its own process; the card shows live progress.
4. `wait_for_run` returns the summary metrics; the model writes the answer from them.

3 Reading a run card

Field	Meaning
order	Initial and final order from $m_1 \cdot m_2$: AFM below −0.9, FM above +0.9, otherwise canted.
m_1, m_2, Néel final	Final unit vectors and the Néel vector $n = (m_1 - m_2)/2$.
n_z switches	Real reversals of n_z (from below −0.5 to above +0.5 or back) and the time of the first; ringing around zero is not counted.
settled after	When m_1 and m_2 stop changing within 2 %; "–" if they did not settle within the run.
dominant f	Main frequency of m_{1x}, reported only if at least five periods fit in the second half of the run.

The **sphere** shows the trajectories of m_1 (blue) and m_2 (magenta) in 3D or as an x–z dial. The **chart** shows m_1/m_2, z-components only, or the Néel vector: drag to zoom, double-click to reset, click to move the playhead; **Play** animates the run. The card links the raw files and exports CSV or a zip of all data.

4 Fast, Deep and next steps

Fast uses the default parameters, changes only what you name and answers in a few lines. **Deep** works from the full parameter set or your file, states the caveats that apply and gives a quantitative explanation. Both end with up to three suggested follow-ups. Requests estimated above `CONFIRM_MINUTES` or `CONFIRM_RUNS` pause for **Run it**.

5 Models and personas

Under the message box you pick a persona, not a model. Each persona groups models of one level from both providers.

Persona	Best for	Typical models
Hermes	Single runs, presets, short answers	Claude Haiku · GPT nano / mini
Dionysus (default)	Everyday runs with a clear explanation, sweeps, comparisons	Claude Sonnet · GPT-5.x
Zeus	Deep mode, goal searches, careful curve reading, reports	Claude Opus · OpenAI o-series
Apollo	New or special models as they appear	Claude Fable / Mythos · new models
Athena	The how-to guide: where things are, what buttons do	Selected provider · built-in answers offline

How the model is chosen

On *Auto* each request is scored for every model of the persona whose key is present:

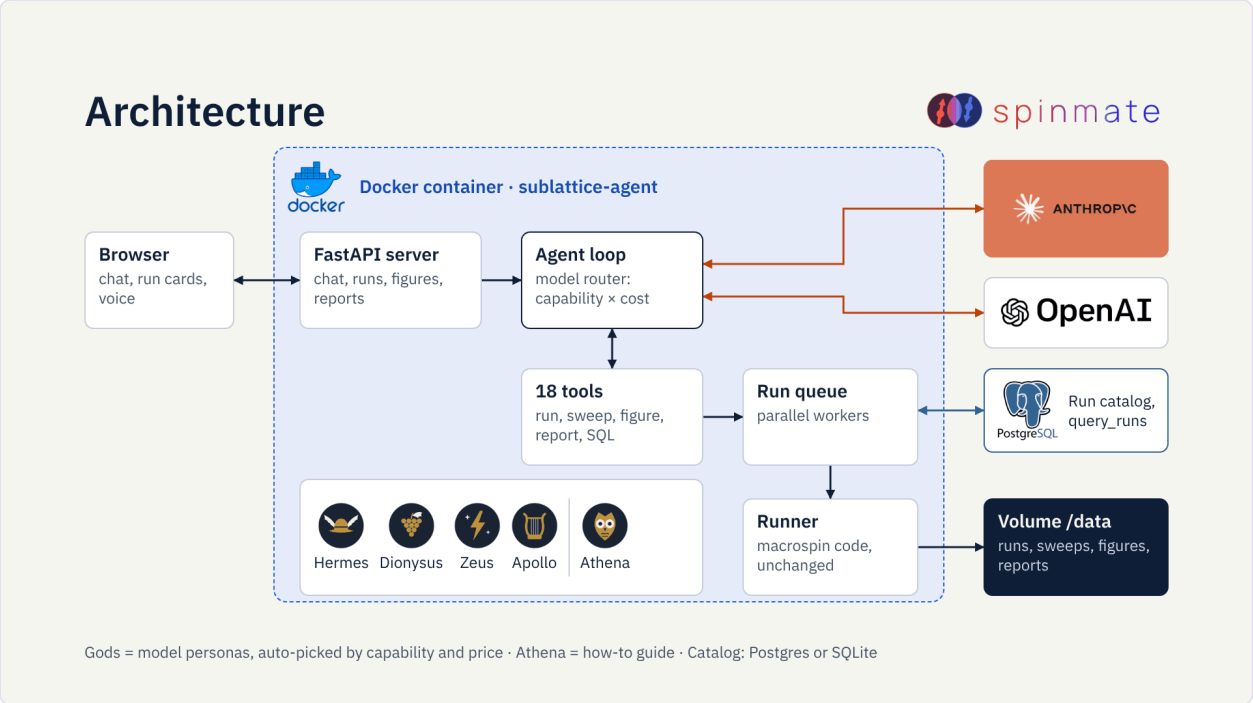
```
score = capability + bonus(strongest of the persona) − λ(persona, request) × log10(price / cheapest) − shortfall
```

Capability comes from the model family and generation; price from list prices; λ is largest for Hermes and smallest for Zeus and shrinks for demanding requests (Deep mode, sweeps, comparisons, reports, long messages). So the strongest model wins unless a cheaper one is clearly enough. There is no fixed provider order; on an API error the next model by score is used. Pro models run only when pinned (Settings → *Model behind each god*). Every answer shows the model, the reason and the cost in €, converted with the daily ECB reference rate.

6 The tools

Group	Tools
Inputs	`list_input_files`, `read_input_file`, `save_input_file`, `validate_parameters`
Runs	`start_simulation`, `wait_for_run`, `get_run`, `list_runs`, `cancel_run`
Sweeps	`start_sweep`, `wait_for_sweep`, `get_sweep`
Continuation	`continue_run`, `combine_runs`
Analysis & output	`compare_runs`, `make_figure`, `make_report`, `query_runs` (SQL on the catalog)

7 Architecture



Component	Role
Browser (`web/index.html`)	Chat, run cards, sphere and charts, Runs panel, figure and report cards, voice input. Conversations are kept in the browser.
FastAPI server (`app/server.py`)	REST endpoints for runs, sweeps, figures, reports, accounts, transcription; streams the agent's events to the page.
Agent loop (`app/agent.py`)	Provider-neutral tool loop for Claude and OpenAI, with prompt caching, token budget and cost accounting.
Model router (`app/router.py`)	Scores the persona's models on capability and price for each request.
Tools (`app/tools.py`)	18 tools the model can call (section 6); large requests return a confirmation instead of starting.
Run queue (`app/simulations.py`)	Parallel workers, sweeps, continuation and combination of runs, summaries.
Runner (`app/runner.py`)	Runs the original `macrospin/` code in its own process and computes the metrics of the run card.
Run catalog (`app/catalog.py`)	Postgres in Docker, SQLite in the desktop app: one row per run, queried with read-only SQL.
Figures, reports	matplotlib/SciencePlots figures; HTML/PDF reports with MathML math and a Markdown zip.
Data volume	`/data`: runs, sweeps, figures, reports, accounts, exchange-rate cache.

8 Sweeps, threshold searches and continuing runs

- **Sweep:** “Sweep A0 from −0.25e-12 to +0.25e-12 in 11 steps, 0.5 ns each.” Runs go in parallel; the card shows a table and an overlay.
- **Threshold search:** “Find the smallest SOT current between 50e6 and 150e6 A/cm² that makes the Néel vector rotate.” A few rounds of bracketing sweeps narrow the interval.

- **Continue:** “Continue r0003 for another 0.5 ns.” Time does not restart; *combine* joins the segments into one curve.

9 Comparing runs, figures and reports

- **Compare:** lists the parameters that differ and overlays the curves.
- **Figures:** “Make an IEEE figure of n_z for r0001 and r0003, first 60 ps.” Styles science, IEEE, Nature; PDF, SVG and PNG downloads; *Change...* to adjust.
- **Reports:** SpinMate header, method, results, parameters, figures and appendix, with LaTeX math; print to PDF or download the Markdown version.

10 The Runs panel and questions across runs

The list icon opens every run: search by id, label, #tag or parameter; filter by status and end state; star and tag; select several to compare, plot, report or delete. With the catalog the assistant answers questions across runs in one read-only query:

> *Make me a table of all finished runs: id, A0, initial and final order, first n_z switch in ps.*

11 Settings and conversations

- **Theme:** Spin (default), Light or Dark. **Default parameters:** saved per user and used as the base of every run.
- **Cost:** tokens and the estimated € next to the mode buttons; hover for the exchange rate.
- **Conversations:** the clock icon lists your chats; select one, several or all and delete them (runs are kept).
- **Voice:** the microphone records a request; the arrow next to it chooses the microphone and language.

12 Data, privacy and security

- The app listens on this computer only (`BIND_ADDRESS=127.0.0.1`). There is no password; expose it only behind a proxy with real authentication.
- API keys stay in `.env` (never committed) or in the browser tab. Calls are billed to your own provider account; set `TOKEN_BUDGET` and a monthly limit in the provider console.
- Accounts (name and email) are stored only in `/data/accounts.json` on your machine; nothing is sent to the authors.
- The assistant reads the catalog with read-only SQL and a timeout, and starts runs only through the solver's parser.

13 Configuration (`.env`)

Variable	Purpose
`ANTHROPIC_API_KEY`, `OPENAI_API_KEY`	Server-side keys (optional; can be pasted in Settings)
`HOST_PORT`, `BIND_ADDRESS`	Port (8421) and who may open the app (127.0.0.1)
`TOKEN_BUDGET`	Token limit per server session
`CONFIRM_MINUTES`, `CONFIRM_RUNS`	When a request waits for **Run it**

`MAX_PARALLEL`, `MAX_SWEEP_RUNS`	Parallel runs and the largest sweep
`DATABASE_URL`, `POSTGRES_PASSWORD`	Run catalog; empty URL switches it off
`EUR_PER_USD`, `PRICES_JSON`	Fixed exchange rate (default: daily ECB) and price overrides
`SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, `SMTP_FROM`	Mail server for the optional welcome email (Gmail: smtp.gmail.com, 587, App Password)
`SSL_CERT_FILE`	Company CA bundle behind a corporate proxy

14 Parameters (most used)

Parameter	Meaning
`--t`	Duration (s)
`--A0`	Inter-sublattice exchange (J/m); negative couples antiparallel
`--m1`, `--m2`	Initial unit vectors
`--SOT_DC_Amp`, `--SOT_pol`	DC spin-orbit current (A/cm²) and its polarisation
`--flagTempVarying`, `--Temp`, `--Fr`	FeRh temperature cycle: on, amplitude (K), frequency (Hz)

A parameter file is plain text, one `--name=value` per line; attach it with the paper clip. Ask “Which parameters can I set?” for the full list with units. Examples are in `examples/`.

```
--t=5e-11
--A0=-0.248e-12
--m1=0.25,0.0,0.968
--m2=0.35,0.0,0.937
```

15 Troubleshooting

Symptom	What to do
“Add an API key”	Paste a key in Settings, or set it in `.env` and run `docker compose up -d --force-recreate`.
No answer, 401 / 429 errors	Check the key and the provider credit; on Auto the next model by score is tried.
New features do not appear	`git pull`, `docker compose up --build -d --force-recreate`, then Ctrl+Shift+R in the browser.
No welcome email	Check `SMTP_*` in `.env`; `/api/config` must show `"email":true`. The guide is always at `/api/guide.pdf`.
Page does not open from another computer	By design; see `BIND_ADDRESS` and section 12.
Microphone not listed	Allow the microphone once; open the app on localhost or https.

More in the README on GitHub. Issues and feedback are welcome.